

the messages shared between the broker and consumer are classified based on the topics that the particular broker in the Kafka cluster holds.

Zookeeper

In order to define topics and keep a metadata record of all the topics defined in the cluster, Kafka uses Zookeeper, which is a replicated distributed log with a file system API built on top. You can think of Zookeeper in Kafka as something similar to YARN in Hadoop. Not only does it keep a record of all the topics within all the nodes in the cluster, it also keeps track of which nodes are down and which are up and running. It also decides which topic should be created in which node in order to bolster efficiency.

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Decentralised platforms are very often associated with blockchain based environments in which complete security and privacy is given to data.

These decentralised platforms can be programmed using various programming toolkits and frameworks including Python, JavaScript, HTTP APIs, PHP and Java. With the integration of a blockchain based environment in the storage repositories and database applications, multi-dimensional security and privacy of data can be achieved. Many open source communities and corporate giants are working towards this.

To sum up, data leakage and identity theft is on the rise due to the billions of connected devices in use today. Traditional database engines, file repositories and storage systems are not secure because of the centralised environment, which makes it easy for hackers to penetrate into the central server and fetch files. With the use of blockchain based decentralised storage platforms, the overall security, integrity and privacy of data files and credentials can be increased effectively. 

Next steps

The tools and frameworks discussed in this article are a high level overview of the innovation and efforts taking place in the Big Data management industry. This is especially true from the open source perspective. More than 40 per cent of the Fortune 500 companies deploy at least one if not all of the frameworks mentioned in this article. But there are many more open source tools and frameworks that can be learnt and used for Big Data management. Another

amazing thing that I have noticed about such frameworks is that they have very active open source communities with merge requests being quickly responded to and forums filled with innovative ideas. This can be a healthy place for you to get started in the open source community. One thing is for sure — adding a Big Data management tool to the toolkit in your resume will certainly make it more attractive. As discussed in the second section of this article, we have the data but lack talented analysts. 

By: Mir H.S. Quadri

A senior content developer at *SimplyPut.fyi*, the author is a research analyst with a specialisation in artificial intelligence and machine learning. He shares a deep love for the analysis of technological trends and understanding their implications. Being a FOSS enthusiast, he has contributed to several open source projects.

Cloud platform	URL
Sia	https://sia.tech/
Storj	https://www.storj.io/
IPFS	https://ipfs.io/
Swarm	https://www.ethswarm.org/
GUN	https://gun.eco/
OrbitDB	https://github.com/orbitdb/orbit-db
FileCoin	https://filecoin.io/
BlockChainDB	https://www.bigchaindb.com/

Table 2: Decentralised platforms

References

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By: Dr Gaurav Kumar, Dr Srinivasa K.G. and Amit Doegar

Dr Gaurav Kumar is the MD of Magma Research and Consultancy. He is associated with various academic institutes, where he delivers expert lectures and conducts technical workshops on the latest technologies and tools.

Dr Srinivasa K.G. is a professor in the Department of Information Management and Emerging Engineering at the National Institute of Technical Teachers Training and Research (NITTTR) in Chandigarh.

Amit Doegar is associate professor in the Department of Computer Science and Engineering, National Institute of Technical Teachers Training & Research (NITTTR), Chandigarh.